



INDIVIDUAL REPORT – DBMS PROJECT

Neel Solanki (GROUP 7)

Neel Solanki

OVERVIEW

Three main activities that addressed the whole data pipeline—from data integration to data analysis—were engaged in this project. I developed a Talend ETL job in Task 1 to load, clean, and extract address data from SQL Server to Oracle while using transformation rules to guarantee data quality. I created a Talend pipeline in Task 2 to import contribution entries from many CSV files, rejecting incorrect values and checking foreign keys with the ADDRESS and VOLUNTEER tables. I created the FactDonations table in Task 3 and assisted in putting a star schema with associated dimension.

Task 1: Address Table Integration using Talend

Objective

Task 1's main goal was to get address entries from a Microsoft SQL Server database and add them to an Oracle SQL table while utilizing Talend Open Studio to fix data discrepancies using transformation logic.

My Contributions & Role

I was in charge of setting up the complete ETL pipeline and the data transformation algorithms as the technical lead. Below is a detailed explanation of my duties:

1. Configuring the database

worked using Oracle SQL Developer to use the following schema to create a table named ADDRESS:

```
CREATE TABLE ADDRESS (  
    Address_ID NUMBER PRIMARY KEY,  
    Unit_Num VARCHAR2(6),  
    Street_Number NUMBER NOT NULL,  
    Street_Name VARCHAR2(24) NOT NULL,  
    Street_Type VARCHAR2(12) NOT NULL,  
    Street_Direction CHAR(1),  
    Postal_Code CHAR(7) NOT NULL,  
    City VARCHAR2(16) NOT NULL,  
    Province CHAR(2) NOT NULL,  
    CONSTRAINT St_Dir CHECK (Street_Direction IN ('E','W','N','S'))
```

);

1. Talend Job Design

- Created a Talend job with the tMap, tDBInput, and tDBOutput components. Developed two metadata database connections: one for Oracle (AddressOracle) and one for SQL Server (U07.World).
- Set up tDBInput to get 70,088 SQL Server records.

2. Error Resolution

- Null values in non-nullable columns caused the initial task to fail.
- Successfully reprocessed the records after applying conditional logic and debugging the problem using Talend's logs and console view.

3. Final Output

- Confirmed that there were no duplicates or constraints in the cleansed address data when it was imported into Oracle.
- The basic dataset for subsequent processes like Fact Table modeling and Donation ETL was supplied by this assignment.

Task 2: Donation Data ETL Pipeline with Validation

Goal

To simulate a real-world data integration challenge, we were tasked with importing donation records from multiple CSV files, ensuring only valid rows (based on foreign key relationships) were inserted into the DONATION table, and rejecting invalid records.

My Role & Contributions

1. Schema Creation

- Designed two Oracle tables:
 - VOLUNTEER – to track volunteer IDs and leader-member relationships.
 - DONATION – to store donation info with foreign keys referencing ADDRESS and VOLUNTEER.

CREATE TABLE VOLUNTEER (

Volunteer_ID NUMBER PRIMARY KEY,

First_Name VARCHAR2(16),

Last_Name VARCHAR2(16),

Group_Leader NUMBER,

```

CONSTRAINT fk_vol_lead FOREIGN KEY (Group_Leader) REFERENCES VOLUNTEER(Volunteer_ID)
);

CREATE TABLE DONATION (
    Donation_ID NUMBER PRIMARY KEY,
    Donor_First_Name VARCHAR2(16),
    Donor_Last_Name VARCHAR2(16),
    Donation_Date DATE NOT NULL,
    Donation_Amount NUMBER(7,1) NOT NULL,
    Address_ID NUMBER NOT NULL,
    Volunteer_ID NUMBER NOT NULL,
    CONSTRAINT fk_don_add FOREIGN KEY (Address_ID) REFERENCES ADDRESS(Address_ID),
    CONSTRAINT fk_don_vol FOREIGN KEY (Volunteer_ID) REFERENCES VOLUNTEER(Volunteer_ID)
);

```

2. Data Preparation

- Created three distinct CSV files (Neel.csv, Pratham.csv, Vimal.csv) with:
 - Valid entries (correct foreign keys).
 - Invalid entries (nulls, unmatched Address_IDs, etc.).

3. Talend Pipeline

- Used tFileList to loop over all CSVs.
- tFileInputDelimited read CSV line-by-line.
- Lookups were performed via tDBInput from VOLUNTEER and ADDRESS tables.
- tMap was used to filter:
 - Invalid Data:


```

row5.Address_ID == null || row7.ADDRESS_ID == null ||
row5.Volunteer_ID == null || row1.VOLUNTEER_ID == null ||
!row5.Address_ID.equals(row7.ADDRESS_ID) ||
!row5.Volunteer_ID.equals(row1.VOLUNTEER_ID)

```

▪ Valid Data:

```
row1.VOLUNTEER_ID != null &&  
row7.ADDRESS_ID != null &&  
row5.Volunteer_ID != null &&  
row5.Address_ID != null &&  
row1.VOLUNTEER_ID.compareTo(row5.Volunteer_ID) == 0 &&  
row7.ADDRESS_ID.compareTo(row5.Address_ID) == 0
```

4. Output Handling

- Valid records: inserted into the DONATION table.
- Invalid records: exported to CSV using `tFileOutputDelimited` and simultaneously logged via `tLogRow`.

5. Result

- Successfully loaded clean data with full integrity.
- Rejected data was reviewed by group members for improvement and debugging.

Task 3: Star Schema & Data Mart Design

Goal

To design and implement a dimensional data mart using a star schema, enabling donation analysis by time, volunteer, and location.

My Role & Contributions

1. Data Mart Design

- I initiated the DATAMART schema connection and led the creation of the central fact table: `FactDonations`.

2. Fact Table:

```
CREATE TABLE FactDonations (
```

```
date_id DATE,
```

```
address_id NUMBER,
```

```
volunteer_id NUMBER,
```

```
donation_count NUMBER,
```

```
total_donation_amount NUMBER(10, 2),  
CONSTRAINT fk_date FOREIGN KEY (date_id) REFERENCES DimDate(date_id),  
CONSTRAINT fk_address FOREIGN KEY (address_id) REFERENCES DimAddress(address_id),  
CONSTRAINT fk_volunteer FOREIGN KEY (volunteer_id) REFERENCES DimVolunteer(volunteer_id)  
);
```

3. Dimension Tables – Created by Me and Collaboratively Refined

- DimDate

```
CREATE TABLE DimDate (  
    date_id DATE PRIMARY KEY,  
    year NUMBER(4),  
    month_number NUMBER(2),  
    day_number_in_month NUMBER(2),  
    month_name_short VARCHAR2(10),  
    month_name_long VARCHAR2(20)  
);
```

- DimAddress

```
CREATE TABLE DimAddress (  
    address_id NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    postal_code VARCHAR2(10),  
    address_line VARCHAR2(100)  
);
```

- DimVolunteer

```
CREATE TABLE DimVolunteer (  
    volunteer_id NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    volunteer_name VARCHAR2(100)  
);
```

4. Schema Integrity Validation

- Ensured the FK relationships were working correctly by inserting test records.

- Assisted group members in mapping ER Diagram based on implemented schema.

5. Result

- Created a query-ready schema structure for OLAP:
 - Donations by month
 - Top performing volunteers
 - Geographical contribution analysis

FINAL CONCLUSION

Throughout the completion of Tasks 1, 2, and 3 in this Database Management project, I gained deep, practical experience in building a full-fledged data integration and warehousing solution.

In **Task 1**, I learned how to extract large-scale data from a relational SQL Server source and successfully migrate it into Oracle while managing data cleaning and transformation using Talend. This task taught me the importance of data quality and how conditional logic within ETL tools ensures robust, error-free database population.

In **Task 2**, I implemented a dynamic and reusable Talend pipeline to handle foreign key validations, multiple CSV imports, and conditional rejections. This not only reinforced my understanding of referential integrity and ETL automation but also gave me hands-on experience with real-world data inconsistencies and exception handling—core skills required for data engineering roles.

In **Task 3**, I designed and validated a Star Schema data mart, involving fact and dimension tables, with the correct level of granularity. I established relationships between time, geography, and volunteer effort, providing a strong foundation for analytical reporting and decision-making. This task emphasized the value of dimensional modeling for building scalable and query-efficient data systems.

Together, these tasks simulated an end-to-end ETL and data warehousing project, from source system extraction to data mart design. They significantly enhanced my technical capabilities in database development, Talend ETL workflows, SQL programming, and analytical schema modeling, while also strengthening my problem-solving and leadership skills in a group project environment.